# THE COMBINATION MITER/ MORTISE-AND-TENON

## A great joint for a variety of applications

TEXT AND PHOTOS BY J. SPEETJENS



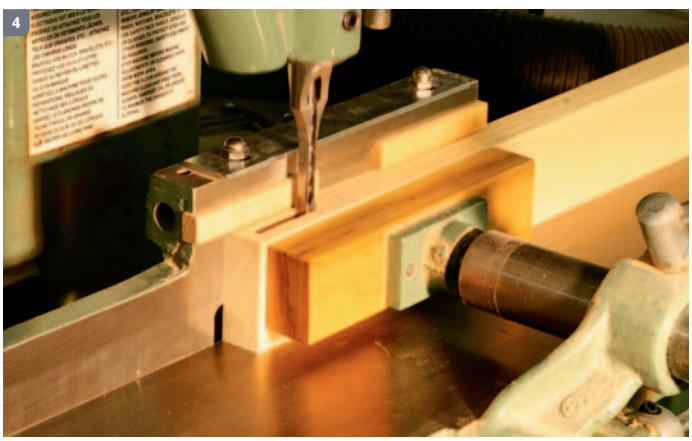
The combination miter/mortise-and-tenon is a strong and versatile joint, and the end result provides a satisfying sense of accomplishment. It may seem at first a little difficult, but once you have made a few simple setups and become familiar with the process, you will be able to efficiently and accurately produce it with ease. The joint can be used as a substitute for the standard cope and pattern sets, or it can be used to create frames with unique detailing. You can use the process to make joints with deep or through tenons, thereby creating doors of substantially added strength and character—something no typical cope and pattern set can achieve. As a professional woodworker, I am frequently asked to make one or two doors to match existing details; with the combination miter/mortise-and-tenon joint and an inexpensive router bit or two, I can replicate details at a fraction of the cost of purchasing specialized tooling. I also use the combination joint to create beaded frames, which result in a clean-looking job that eliminates the fuss of making, fitting, and nailing lots of auxiliary beading.

The basic joint is made up of two components: a mortised stile and a tenoned rail, each of which has a molded edge that is mitered at the interface. Stiles (the vertical members of a door or frame) usually run the full length of the assembly and the rails typically fit between the stiles. The process and techniques that follow apply to frames that have a relatively narrow molded edge, typically less than 25% of the stile or rail width.


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When you are making up your frame stock, mill a couple of extra feet for test cuts. Set up and cut a sample of the molded profile (1). The width of the molded detail is used to determine the size of the tenon's shoulder. For all but the narrowest of rails, I line up the interior end of the mortise with the long point of the miter. When I cut the associated tenon I allow for 1/32" clearance, thereby helping to ensure a tight-fitting miter. For rails that have two molded edges and consequently two miters, this leeway will allow the miters to automatically center the rail during assembly. For single-edged rails, if there are no other considerations I will make the second shoulder the same size as the first.

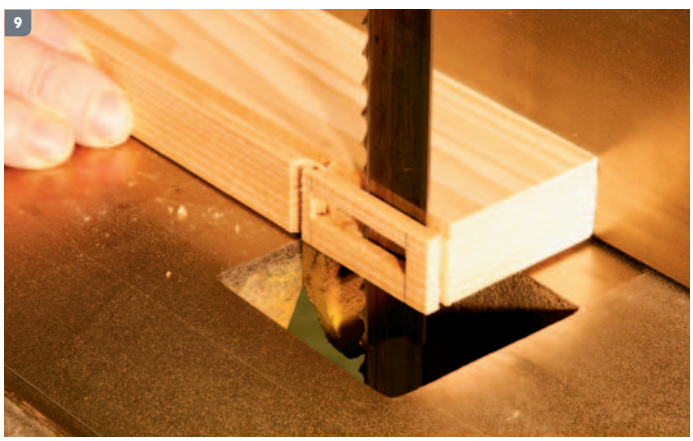
Start by balancing the rail flush on the end of the stiles (2) or the appropriate distance off the end for intermediate rails. Use a square and a pencil to mark off the width of the rail. Using the width of the molded profile, lay out the location and length of the mortise. A marking knife makes a precise and un-muddled line (3).

The nature of my business affords me the luxury of an industrial hollow chisel mortiser (4), but there are numerous bench-top models available that will serve nicely. Alternatively, chopping mortises by hand is not so difficult, and the experience is enjoyable. Regardless of the system you use, cut the mortises clean to the mark and allow an extra 1/8" in depth for wood movement across the width of the stile. If the mortise is too shallow, normal seasonal shrinkage can pull the joint open.

Transfer the line that denotes the edge of the rail to the back face of the stile (5). Angle the line to indicate the direction of the miter. Miters angling one way will be marked off on the right stile; miters angling


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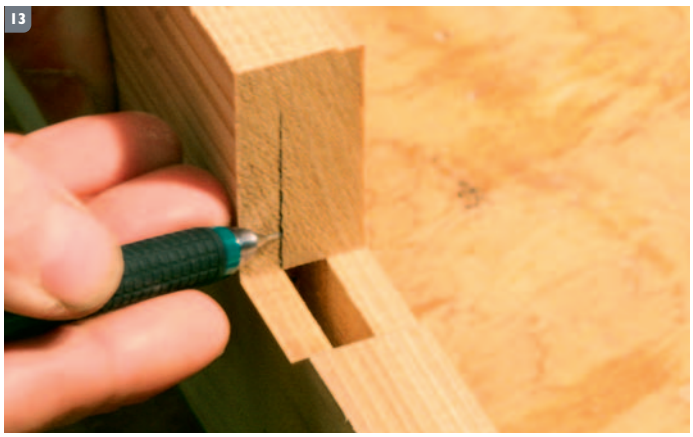

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the opposite way are marked off on the left. Miters on the rails are laid out according to the amount that you have designated for the length of your tenon plus the width of the molded detail. For through tenons (**6**), this is equal to the width of the stile. When determining how long to make your rails, it is important to remember that the effective length of the tenon is reduced by the width of the molding.

The miters are cut into the edge of the frame with a crosscut setup on the tablesaw. I attach a wooden crosscut bar to my sliding table, but a sled can be made that will do the job just as nicely. Set the depth of the miter on a piece of scrap stock. Line the angled mark up with the saw teeth and clamp a stop block to your crosscut bar or the backstop of your sled. I try to set the stop so that I can make a test cut just to the waste side of the line. I make a short nibble cut (**7**), check my result, and adjust the stop block accordingly with a gentle tap of a hammer. Note that consistent rail width is essential for consistent results of intermediate or parting rails. It is also best practice to set your stop at the same end of the stile for all miter cuts in order to eliminate any discrepancies that could arise from variations in the length of the stiles or rails. Cut the miters with a single, even motion so that the depth of cut stops just shy of the molding line (**8**).

Most molding details yield a clear line that can be used to delineate the seat for the tenon shoulders. Set up a fence on the bandsaw to cut about 1/32" to the waste side of this mark and just shy of the miter (**9**). Trim to the line using a bearing-guided flush trim bit in a router table. Some molding details provide a good surface that can guide the bearing of a pattern bit; if not, set














up a fence (10). Use a chisel to clean up the corners at the long point of the miters (11).

If you have the space and tools to set up the molding cut, chop your mortises and cut the miters, then do all these operations before you actually cut the molding detail on your stock. This will ensure "blowout-free" miters; if not, make sure your tools are sharp and clean.

At this point, your stiles are complete with the molded edge, mortise, miter, and seat for the rail shoulders, and the rails are milled with the molding and miters. I like to cut tenons on the tablesaw with a dado set (12). This setup is quick and easy and allows you to work to a precise fit. The process is similar to cutting the miters; again, use a crosscut cradle or arm with a wooden backstop. However, there is a caution about this method: because you are indexing the thickness of the tenon off the two faces, any variations in your stock thickness will translate into variations in tenon thickness. Use a stile with a mortise to transfer a mark defining the thickness of the tenon (13). This line is used for general setup. The point where the miter intersects the edge of the molding determines the length of the tenon. Cut a tenon and check your fit (14). Dry tenons should slide into place without any play. If the fit is too tight, the swelling that occurs with water-based glues will make joint alignment and frame assembly difficult. Depending on your stock thickness and the width of the molded detail, you will probably have to raise the dado blade to cut the shoulders to their full width (15).

For house doors or for panel assemblies with wide rails that have the potential for significant movement, you can use a variation on the above procedure to manually


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cut a cope on the rail member. Notch the molding on the stile at 90° and cut the miter on the rail as above. With carving tools, use the profile of the miter cut to pare away the cope.

In the case of paneled doors and frame, you will have to mill a dado to receive the panel (**16**). For this project I cut stop dadoes on the tablesaw. Alternately, you can run the groove the full length of the stile, but depending on the molding detail and groove depth you may have to incorporate this element into the design of the tenon. Typically, if I am making multiple doors or panels, I will run the groove on through and create the more complex tenon, but for one or two the stop dado is faster. When making stop dadoes, be sure to have clear starting and stopping points marked on the saw's throat plate. Also, be sure to lower the work onto the cutter slowly and have the opposite end held firmly to the table. Keep the stock moving and lift it off the cutter, making sure to keep your hands safely away from the cutter (**17**). **Caution! Never have your hands on the workpiece on the outfeed side of the saw blade.**

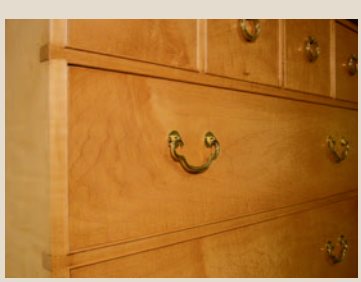
As with any joining operation, the accuracy of your work can either be realized or lost in assembly. Once your frame is glued and clamped, check the length of the diagonals (**18**)—if they are equal, the frame is square; if not, shift the head of one or more clamps 1/2"-1" in the direction that will cause the clamp to compress the longer diagonal. Recheck your diagonals and adjust your clamps as needed. In general, the more accurately you have cut your joints, the easier the assembly will be.

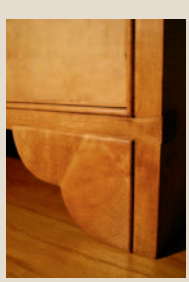
Stand the clamped door or frame on edge, making sure there are no stresses distorting the assembly. Sight down across two parallel members and determine if the frame is flat or twisted (**19**). If the frame is twisted, gently rack it in the opposite direction and re-sight the frame. When it naturally maintains flatness, set the frame off to the side. I re-sight the frame again or use an electronic level to verify flatness (**20**). Winding sticks made from two contrasting colors can be a useful visual aid when sighting for flatness (**21**). Hint: creating a twist-free work surface will make your clamped-up assemblies flatter from the start.

As with most projects the accuracy of each step affects subsequent ones. If you work cleanly and methodically, making sure that your crosscut setups are cutting square, the system of stops will ensure consistently fitting joints. Remember to make sample cuts and, if you are working with large batches, occasionally check your work to verify that you setups haven't changed. I have been using the combination miter/mortise-and-tenon joint regularly for fifteen years and continue to find new applications for it, and I continue to enjoy the feel of a well-fitted joint sliding into place.

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## Miter/Dovetail Joint





The combination miter/dovetail joint (two versions at right) is similar in use and in the methods of fabrication as the combination miter/mortise-and-tenon joint discussed in the article. The example pictured here is used to connect parting rails to the solid sides of a chest of drawers. As with the mortise version, the miter is defined by the molded detail; in this case, a 1/8" raised cock bead. The tails were cut using a tenoning jig on the tablesaw, and the sockets were cut and individually fitted by hand. The process is unquestionably slower than biscuits or dados, but there is little doubt in my mind that barring flood, fire or the wrath of the landfill, a case made with the combination miter/dovetail joint will be around for centuries to come.